Practical - 1

Problem Definition

• Create a React application using Vite for a student portfolio page.

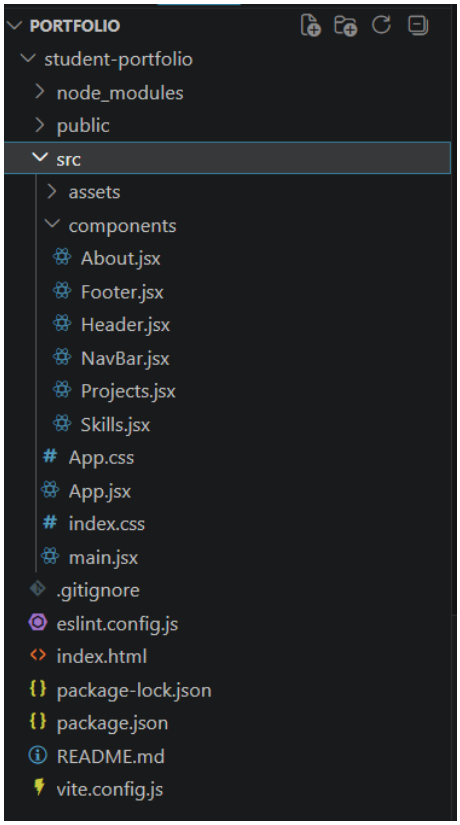
• The application must include at least 4 reusable components: Header, About, Skills, and Footer.

• Each component must be independently structured and composed into a single-page layout.

• No logic or JSX should be duplicated across components.

• Props must be used to pass at least one piece of data into a minimum of 2 components.

Folder Structure:



Steps:

1. Verify Node.js and npm are installed:

```
PS C:\Users\Bhumi> node -v
v24.18.0
PS C:\Users\Bhumi> npm -v
11.18.0
PS C:\Users\Bhumi>
```

2. Scaffold the React app using Vite:

```
PS C:\Users\Bhumi>  npm create vite@latest student-portfolio
Need to install the following packages:
create-vite@9.1.2
Ok to proceed? (y) y

> hello@1.0.0 npx
> create-vite student-portfolio
```

Main.jsx

```jsx
import React from 'react'
import ReactDOM from 'react-dom/client'
import App from './App.jsx'
import './index.css'

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>,
)
```

3. Inside src/components, create four files: Header.jsx, About.jsx, Skills.jsx, Footer.jsx.

Header.jsx

```jsx
function Header({ name, tagline }) {
  return (
    <header className="hero">
      <p className="hero-eyebrow">{name} &mdash; Portfolio</p>
      <h1 className="hero-title">Hi, I'm {name}. {tagline}</h1>
      <p className="hero-subtitle">
        IT student building full-stack web applications
        with React, Node.js, and MongoDB.
      </p>
      <div className="hero-actions">
        <a className="btn btn-primary" href="#projects">View My Work</a>
        <a className="btn btn-secondary" href="#contact">Contact Me</a>
      </div>
    </header>
  )
}

```

```
23.export default Header
```

About.jsx

```
2. function About() {
3.   return (
4.     <div className="bento-card span-3" id="about">
5.       <p className="card-number">01</p>
6.       <h2 className="card-title">About Me</h2>
7.       <p className="card-text">
8.         I am an Information Technology student passionate about
   creating
9.         modern, responsive, and user-friendly web applications. I enjoy
10.        learning new technologies, solving real-world problems, and
11.        developing projects that enhance my skills in web development.
12.      </p>
13.    </div>
14.  )
15.}
16.
17.export default About
```

Skills.jsx

```
3. function Skills({ skillList }) {
4.   return (
5.     <div className="bento-card span-3" id="skills">
6.       <p className="card-number">05</p>
7.       <h2 className="card-title">Skills</h2>
8.       <div className="skills-groups">
9.         {skillList.map((group) => (
10.          <div key={group.category}>
11.            <p className="skill-group-title">
12.              <span className="skill-dot" style={{ background:
   group.color }}></span>
13.              {group.category}
14.            </p>
15.            <div className="skill-tags">
16.              {group.skills.map((skill) => <span className="skill-tag"
   key={skill}>{skill}</span>)}
17.            </div>
18.          </div>
19.        ))}
20.      </div>
21.    </div>
22.  )
23.}
```

```
24.
25.export default Skills
```

Footer.jsx

```
10.function Footer() {
11.  return (
12.    <div className="bento-card span-3 contact-card" id="contact">
13.      <p className="card-number">06</p>
14.      <h2 className="card-title">Let's build something together</h2>
15.      <p className="card-text">Open to internships, freelance work, and
   collaboration.</p>
16.      <a className="btn btn-secondary"
   href="mailto:student@example.com">Email Me</a>
17.    </div>
18.  )
19.}
20.
21.export default Footer
```

4. Import and compose all four components inside App.jsx in a single layout.

App.jsx

```
5. import Header from './components/Header'
6. import NavBar from './components/NavBar'
7. import About from './components/About'
8. import Projects from './components/Projects'
9. import Skills from './components/Skills'
10.import Footer from './components/Footer'
11.
12.function App() {
13.  const projectList = [
14.    {
15.      title: 'Personal Expense Tracker (UI/UX)',
16.      description: 'Designed a user-friendly interface and wireframes
   focused on a clean and intuitive user experience.',
17.      tech: ['Figma'],
18.    },
19.
20.    {
21.      title: 'Beauties & Bows \u2013 E-commerce Website',
22.      description: 'Developed a responsive front-end website for a
   fashion brand, implementing product display and user-friendly
   navigation.',
23.      tech: ['HTML', 'CSS', 'JavaScript'],
```

```
    },
    {
      title: 'Note-Taking Web Application',
      description: 'Created a web app to add, edit, and delete notes, with a backend integrated with a database for data management.',
      tech: ['PHP', 'MySQL'],
    },
    {
      title: 'Shake & Bake Website',
      description: 'Developed and customized a business website using WordPress, managing layout, design, and content.',
      tech: ['WordPress'],
    },
  ]

  const skillList = [
    { category: 'Frontend', color: '#1d3557', skills: ['React', 'JavaScript', 'HTML', 'CSS'] },
    { category: 'Backend', color: '#457b9d', skills: ['Node.js', 'Express', 'REST APIs'] },
    { category: 'Database', color: '#2a9d8f', skills: ['MongoDB', 'Mongoose', 'MySQL'] },
    { category: 'Tools & Practices', color: '#6c757d', skills: ['Git', 'GitHub'] },
  ]

  return (
    <div className="page">
      <NavBar />
      <Header name="Bhumi Vora" tagline="I build web apps." />
      <div className="bento-grid">
        <About />
        <Projects projectList={projectList} />
        <Skills skillList={skillList} />
        <Footer />
      </div>
      <p className="footer-note">&copy; {new Date().getFullYear()} &middot; Built with React &amp; Vite</p>
    </div>
  )
}

export default App
```

Supplementary Problem:

1. **Add a NavBar component that highlights the active section.**

NavBar.jsx

```
4. import { useState } from 'react'
5.
6. function NavBar() {
7.   const sections = ['About', 'Projects', 'Skills', 'Contact']
8.   const [active, setActive] = useState('About')
9.
10.  return (
11.    <nav className="navbar">
12.      <span className="navbar-brand">Portfolio</span>
13.      <div className="navbar-links">
14.        {sections.map((section) => <a key={section} href={'#' +
   section.toLowerCase()} onClick={() => setActive(section)}
   className={active === section ? 'active' : ''}>{section}</a>)}
15.      </div>
16.    </nav>
17.  )
18.}
19.
20.export default NavBar
```

Projects.jsx

```
7. function Projects({ projectList }) {
8.   return (
9.     <>
10.      {projectList.map((project, index) => (
11.        <div className="bento-card" key={project.title} id={index === 0
   ? 'projects' : undefined}>
12.          <p className="card-number">{String(index + 2).padStart(2,
   '0')}</p>
13.          <h3 className="card-title">{project.title}</h3>
14.          <p className="card-text">{project.description}</p>
15.          <div className="card-tech">
16.            {project.tech.map((t) => <span className="tech-tag"
   key={t}>{t}</span>)}
17.          </div>
18.        </div>
19.      ))}
20.    </>
21.  )
22.}
23.
24.export default Projects
```

7. Run the app and verify all components render correctly without console errors.

Output:

